

Lab 3 - Routing Protocol

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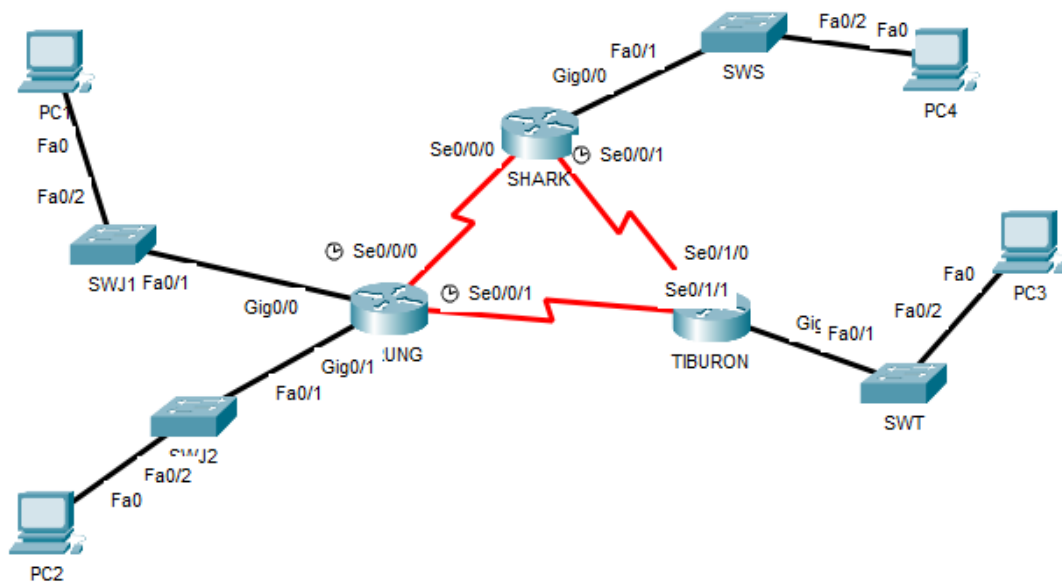
MATRIC NUM: A23CS0227

SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.1	255.255.255.224	-
4		G0/1	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

- Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.

JERUNG LAN1 (20 hosts): /27 (30 usable IPs) → 172.18.110.0/27 (172.18.110.1 – 172.18.110.30) $2^5 = 32$

Subnet mask: /27

Usable IPs: $32 - 2 = 30$

JERUNG LAN2 (30 hosts): /27 (30 usable IPs) → 172.18.110.32/27 (172.18.110.33 – 172.18.110.62) $2^5 = 32$

Subnet mask: /27

Usable IPs: $32 - 2 = 30$

SHARK LAN (60 hosts): /26 (62 usable IPs) → 172.18.110.64/26 (172.18.110.65 – 172.18.110.126) $2^6 = 64$

Subnet mask: /26

Usable IPs: $64 - 2 = 62$

TIBURON LAN (50 hosts): /26 (62 usable IPs) → 172.18.110.128/26 (172.18.110.129 – 172.18.110.190) $2^6 = 64$

Subnet mask: /26

Usable IPs: $64 - 2 = 62$

Connections: Point-to-point links need only 2 IPs each (/30) - JERUNG-SHARK (172.18.110.193 – 172.18.110.194)

- JERUNG-TIBURON (172.18.110.197 – 172.18.110.198)

- SHARK-TIBURON (172.18.110.201 – 172.18.110.202) $2^2 = 4$

Subnet mask: /30

Usable IPs: $4 - 2 = 2$

Device	Subnetwork	Usable Hosts	Total IPs	Usable IPs	Prefix
JERUNG	LAN1	20	$2^5 = 32$	30	$32 - 2 = /27$
	LAN2	30	$2^5 = 32$	30	$32 - 2 = /27$
SHARK	LAN	60	$2^6 = 64$	62	$64 - 2 = /26$
TIBURON	LAN	50	$2^6 = 64$	62	$64 - 2 = /26$
Connections	JERUNG-SHARK	2	$2^2 = 4$	2	$4 - 2 = /30$
	JERUNG-TIBURON	2	$2^2 = 4$	2	$4 - 2 = /30$
	SHARK-TIBURON	2	$2^2 = 4$	2	$4 - 2 = /30$

Device	Subnetwork	Total IPs	Usable IPs
JERUNG	LAN1	172.18.110.0 – 172.18.110.31	172.18.110.1 – 172.18.0.30
	LAN2	172.18.110.32 – 172.18.110.63	172.18.110.33 – 172.18.0.62
SHARK	LAN	172.18.110.64 – 172.18.110.127	172.18.110.65 – 172.18.110.126
TIBURON	LAN	172.18.110.128 – 172.18.110.191	172.168.110.129 – 172.18.110.190
Connections	JERUNG-SHARK	172.18.110.192 – 172.18.110.195	172.18.110.193 – 172.18.110.194
	JERUNG-TIBURON	172.18.110.196 – 172.18.110.199	172.18.110.197 – 172.18.110.198
	SHARK-TIBURON	172.18.110.200 – 172.18.110.203	172.18.110.201 – 172.18.110.202

Device	Subnetwork	Prefix	Subnet Mask
JERUNG	LAN1	$32 - 5 = /27$	11111111.11111111.11111111.11100000 = 255.255.255.224
	LAN2	$32 - 5 = /27$	11111111.11111111.11111111.11111100 = 255.255.255.252
SHARK	LAN	$32 - 6 = /26$	11111111.11111111.11111111.11000000 = 255.255.255.192
TIBURON	LAN	$32 - 6 = /26$	11111111.11111111.11111111.11000000 = 255.255.255.192
Connections	JERUNG-SHARK	$32 - 2 = /30$	11111111.11111111.11111111.11111100 = 255.255.255.252
	JERUNG-TIBURON	$32 - 2 = /30$	11111111.11111111.11111111.11111100 = 255.255.255.252
	SHARK-TIBURON	$32 - 2 = /30$	11111111.11111111.11111111.11111100 = 255.255.255.252

- Using the IP addresses calculated, configure the devices with the appropriate information.

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.

- a. CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

- b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

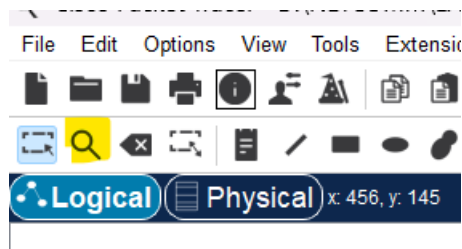


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG

(CLI)

```
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C    172.18.110.0/27 is directly connected, GigabitEthernet0/0
L    172.18.110.1/32 is directly connected, GigabitEthernet0/0
C    172.18.110.32/27 is directly connected, GigabitEthernet0/1
L    172.18.110.33/32 is directly connected, GigabitEthernet0/1
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.193/32 is directly connected, Serial0/0/0
C    172.18.110.196/30 is directly connected, Serial0/0/1
L    172.18.110.197/32 is directly connected, Serial0/0/1
```

(PT Tool)

Routing Table for JERUNG					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0	
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0	
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.193/32	Serial0/0/0	---	0/0	
C	172.18.110.196/30	Serial0/0/1	---	0/0	
L	172.18.110.197/32	Serial0/0/1	---	0/0	

TIBURON

(CLI)

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
```

(PT Tool)

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

SHARK

(CLI)

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C    172.18.110.64/26 is directly connected, GigabitEthernet0/0
L    172.18.110.65/32 is directly connected, GigabitEthernet0/0
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.194/32 is directly connected, Serial0/0/0
C    172.18.110.200/30 is directly connected, Serial0/0/1
L    172.18.110.201/32 is directly connected, Serial0/0/1
```

(PT Tool)

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time=3ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.

PC1 and PC2 ping successfully because they are on the same subnet.

PC1 and PC4 are on different subnets causing they cannot communicate directly to each other. They are also connected to different routers, routing is needed to forward packets between the networks. The router will send 'Destination host as unreachable' when it is not configured properly.

5. What needs to be done to ensure all PCs can ping each other successfully?

-Check the physical connectivity and make sure that all cables and connections between devices are correct and properly established such as switches and routers.

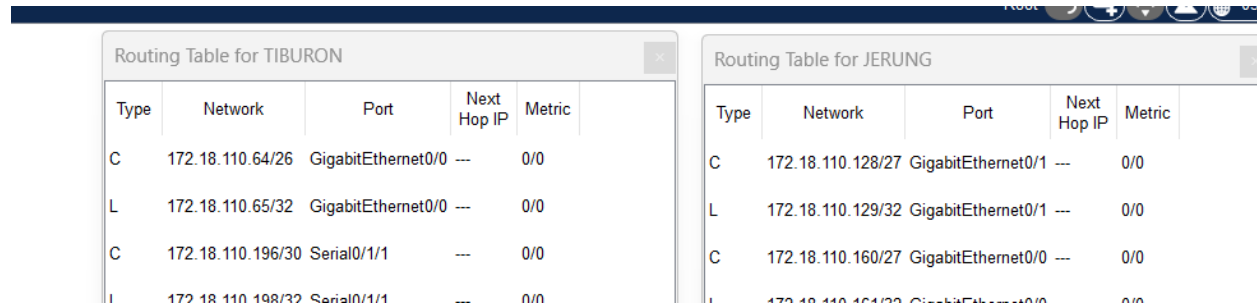
-Check the neighbouring device configurations are correctly configured

We need to confirm that the serial interfaces on the routers are configured correctly with IP addresses, subnet masks

-Lastly, verifying default Gateway on PCs to ensure that all PCs have the correct default gateway configured to send packets to other subnets via the router.

Task 3 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.



Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
I	172.18.110.198/32	Serial0/1/1	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
I	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- What can you say about the addresses used in the 'network' instructions in Figure E?

The 'network' instructions in Figure E tell RIP to advertise the routes of these subnets and include all directly connected interfaces that fall within the specific subnets.

network 172.18.110.128 – include all interface within 172.18.110.128 subnet

network 172.18.110.160 – include all interface within 172.18.110.160 subnet

network 172.18.110.192 – include all interface within 172.18.110.192 subnet

network 172.18.110.196 – include all interface within 172.18.110.196 subnet

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

- a. What are the changes seen in TIBURON?

In the new routing table, TIBURON has new RIP-learned routes (R type) for 172.18.110.0/27 and 172.18.110.32/27, both via 172.18.110.197 on Serial0/1/1, with a metric of 120. The connected (C) and local (L) routes for networks like 172.18.110.128/26, 172.18.110.196/30, and 172.18.110.200/30 remain unchanged. These routes are learned from another router via RIP and are now part of the TIBURON routing table, allowing it to forward traffic to these subnets dynamically.

- b. What are the Networks with type R in TIBURON and JERUNG?

TIBURON

172.18.110.0/27 via 172.18.110.197
 172.18.110.32/27 via 172.18.110.197
 172.18.110.192/30 via 172.18.110.197

JERUNG:

172.18.110.128/26 via 172.18.110.198
 172.18.110.200/30 via 172.18.110.198

- c. Ping PC3 from PC1. Was it successful?

Yes

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=13ms TTL=126
Reply from 172.18.110.190: bytes=32 time=14ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=14ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 10ms
```

- d. Ping PC4 from PC1. Was it successful?

No

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- e. Explain the reasons for your answer.

For PC1 connected to the JERUNG router, and PC3 connected to

The RIP is configured on TIBURON router; hence, their routers have routes of each PCs in their routing table, and this enables them to send the packet to the correct path via refer to routing table.

However, the router SHARK wasn't configured with RIP. This makes PC1 not know the destination of PC4 because of the missing path in the routing table for PC4, which was connected with router SHARK.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.128
SHARK(config-router)#network 172.18.110.160
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.196
SHARK(config-router)#no auto-summary
SHARK(config-router)#exit
SHARK(config)#exit
SHARK#
%SYS-5-CONFIG_I: Configured from console by console
```

- g. Open router SHARK's routing table and paste here.

Routing Table for SHARK				
Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least **twice** to get best results.
- i.

PC	Ping result
1	<pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=14ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=2ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 14ms, Average = 4ms C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=9ms TTL=126 Reply from 172.18.110.30: bytes=32 time=17ms TTL=126 Reply from 172.18.110.30: bytes=32 time=7ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 17ms, Average = 8ms </pre>

2	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=7ms TTL=126 Reply from 172.18.110.62: bytes=32 time=6ms TTL=126 Reply from 172.18.110.62: bytes=32 time=7ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 7ms, Average = 5ms C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=12ms TTL=126 Reply from 172.18.110.62: bytes=32 time=2ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 12ms, Average = 4ms </pre>
3	<pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=7ms TTL=126 Reply from 172.18.110.190: bytes=32 time=12ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 12ms, Average = 5ms C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=10ms TTL=126 Reply from 172.18.110.190: bytes=32 time=5ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 10ms, Average = 4ms </pre>

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: [192.168.1.0/24](#)
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.2
Subnet Mask	255.255.255.54
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?
 No, because they are on different gateway.
 Ping PC3 from PC4

```
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Ping PC3 from PC1

```
Pinging 198.162.1.2 with 32 bytes of data:
Reply from 172.18.110.1: Destination host unreachable.
Request timed out.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 198.162.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1	R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0	L	172.18.110.198/32	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	C	172.18.110.200/30	Serial0/1/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	L	172.18.110.202/32	Serial0/1/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0	C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0	L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

- e. Referring to the routing table, explain your findings.
 TIBURON is directly connected to the 192.168.1.0/24 subnet and routes traffic for other subnets, including those in the 172.18.110.0/24.
 SHARK is directly connected to 172.18.110.64/26 and has the routing information. Respectively for 172.18.110.0/27 and 172.18.110.32/27 subnets.
 Both routers rely on RIP, which shows R (RIP-learn) in the routing table.
- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
 Configure RIP on all routers and include all necessary networks. Next, check for any subnetting that might cause IP conflicts. Use the ping command to test connectivity to and from each PC. If some networks are not reachable, configure either default or static routes as necessary.

- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#network 172.18.110.0
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#exit
TIBURON(config)#exit
TIBURON#
%SYS-5-CONFIG_I: Configured from console by console
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	<pre>C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Request timed out. Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=13ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 13ms, Average = 5ms C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms</pre>

2	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Request timed out. Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms </pre>
4	<pre> C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Request timed out. Reply from 172.18.110.126: bytes=32 time=2ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=2ms TTL=126 Reply from 172.18.110.126: bytes=32 time=2ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms </pre>

REFLECTION

What have you learned in this task?

I have learned to configure IP addresses, subnet masks, and routing on routers. I also learned how RIP works on the routers for them to communicate with each other and therefore all devices connected to different subnets. I also realized why putting the proper gateway on a PC is important for communication to take place and saw how changing one router's IP address may affect an entire network. I can learn how to interpret routing tables and make use of tools such as the CLI and ping commands to verify that the network will work.